A STATISTICAL INVESTIGATION OF ATMOSPHERIC CO_2 AND CH_4 CONCENTRATIONS AT CAPE GRIM OVER FOUR DECADES

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Abstract

As atmospheric gases approach unprecedented levels, analysing their trajectories is key to quantifying the impact of climate change mitigation. This research investigated a 40-year historical dataset of carbon dioxide (CO_2) and methane (CH_4) concentrations from Cape Grim Baseline Station to identify their long-term growth rates and develop a predictive model for the upcoming decade (2026-2035). Systematically, the data was profiled to confirm its reliability, the relationship between both the gases was mapped, historical trends of the datasets were evaluated and future trajectories were projected. The analysis revealed a strong correlation between CO_2 and CH_4. Historically, CO_2 growth rates have accelerated significantly compared to CH_4, which was much more stable. Furthermore, the projection for the 2026-2035 decade exhibited continuous accumulation for both, with CO_2 demonstrating a highly confident, upward trend. Conversely, CH_4 showed a greater predictive uncertainty. Ultimately, this indicates that CO_2 concentrations are actively accelerating, while CH_4 concentrations are increasing at a consistent pace. This varying growth shows that industrial expansion is significantly more responsible for enhancing the greenhouse effect than agricultural factors. Therefore, mitigation strategies must be implemented accordingly.

***Keywords*:** greenhouse gases, carbon dioxide, methane, Cape Grim, predictive model

GitHub Link: https://github.com/PeterNLee/AU_EDA_csiro-co2-ch4-40yr-trend

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1.0 Introduction

Human activities such as industrial expansion, and agricultural intensification have altered Earth's atmosphere by increasing greenhouse gas emissions, specifically, carbon dioxide (CO_2) and methane (CH_4). This has resulted in irregular weather patterns and long-term climate disruptions (Mose & Kinuthia, 2025; Zevenhoven, 2015). While the roles of these gases are understood, their precise observational changes and correlation remain insufficiently defined. Long-term data is essential to investigate past trends and develop reliable predictive models to forecast future atmospheric trends (Jones et al., 2023).

This study examines 40 years of atmospheric recordings between 1986 and 2025 at Cape Grim Baseline Air Pollution Station, Tasmania, Australia. A premier Southern hemisphere source for baseline greenhouse gases (GHG) data. The research aims to quantify atmospheric concentration changes of CO_2 and CH_4, evaluate their correlation, and significant changes in decadal growth. Utilising a statistical framework of univariate analysis, bivariate analysis, and ANOVA testing, the results would assist in developing a predictive model to forecast concentrations of these GHG for the upcoming decade (2026-2035).

2.0 Dataset and Preprocessing

The dataset in this study encompasses 40 years of atmospheric concentration recording for CO_2 and CH_4. The data was acquired from Commonwealth Scientific and Industrial Research Organisation (CSIRO) database (Cape Grim Baseline Air Pollution Station, 2025). Concentration levels were recorded using in-situ chromatography and automated flask sampling at Cape Grim station location in Tasmania, Australia. As observed in Figure 1.0, due to the positioning of the station, it serves as an ideal source for baseline of Southern Hemisphere GHG. The variables being investigated in this study are illustrated in Table 1.0.

Table 1.0: Summary of Greenhouse gas variables and their units of measurements

Greenhouse gas	Units of Concentration
Carbon dioxide (CO_2)	Parts per million (ppm)
Methane (CH_4)	Parts per billion (ppb)

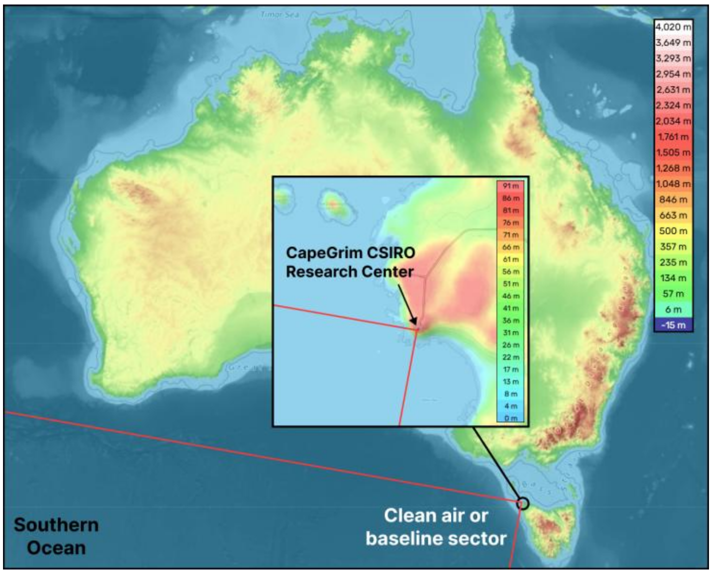

Figure 1.0: Cape Grim CSIRO Atmospheric Research Centre location

To ensure statistical reliability, raw CSIRO dataset was refined. Initially, null values were removed using R programming piping commands. Furthermore, separate Excel files of CO_2 and CH_4 were merged and aligned on yearly basis for performing analysis

3.0 Methodology

3.1 Workflow

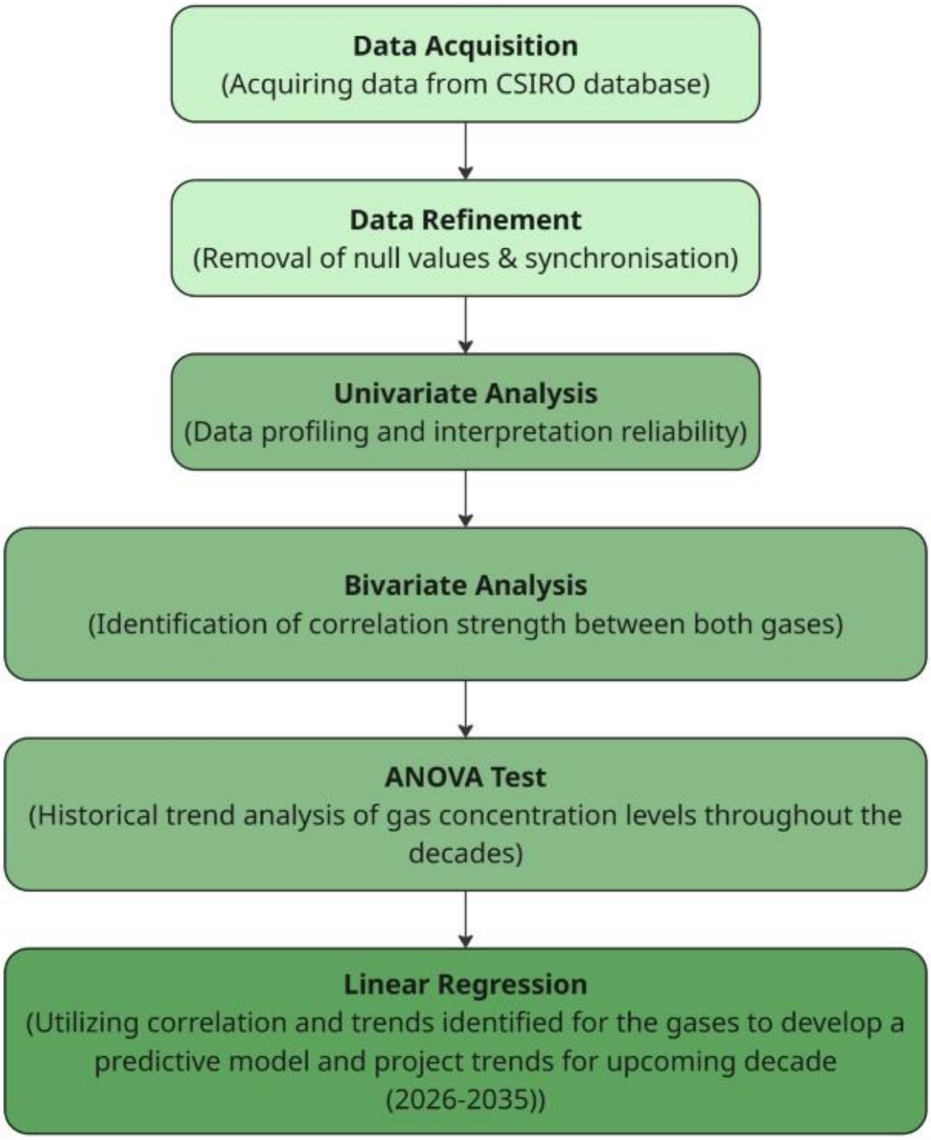

Figure 2.0: Flowchart illustrating steps undertaken to execute the research project

In Figure 2.0, the methodology follows a workflow from raw data collection to future trend forecasting. CO_2 and CH_4 concentrations were obtained from the CSIRO database, and R programming was used to remove null values. Thorough pre-processing ensured both variables were aligned and reliably prepared for statistical evaluation.

3.2 Statistical Profiling and Correlation Analysis

To develop a dataset profile prior to applying comprehensive analytical tools, univariate analysis was performed upon the refined dataset. The five-number summary was computed for both CO_2 and CH_4 concentration levels. Furthermore, histogram and boxplot were generated to visually analyse the data's spread, shape and potential outliers. This initial method was chosen to isolate distribution of each variable, monitor skewness and identify abnormalities, that could impact the statistical evaluation.

After baseline data reliability was confirmed, bivariate analysis was executed. Scatter plots and Pearson's correlation coefficient (r) were generated using R, to assess the relationship between CO_2 and CH_4. This analysed the direction, linearity and strength of correlation. Applying this tool addressed the study's aim, to investigate whether both GHG have accumulated simultaneously over the decades from shared drivers.

3.3 Historical Trend Evaluation

To determine statistically significant changes in CO_2 and CH_4 concentration levels, ANOVA test was implemented. The analyses evaluated the mean growth rates across four decades. Namely, 1986-1995,1996-2005,2006-2015, and 2016-2025.

Prior to executing, assumptions of data's normality, constant variance and independence of variables were validated to ensure reliable results. Hypotheses were formally established; the null hypothesis (H_o) presumed equal means across decades, while alternate hypothesis (H_a) stated that at least one decadal mean differed. Decadal boxplots were generated to visualize historical trends. Finally, a Tukey Honestly Significant Difference (HSD) test was generated, to evaluate statistical significance for the 40-year timeline of concentrations, analysing the calculated p-values against a 5% significance level to support either H_o or H_a.

This specific method was selected to identify if concentration variations were significant enough to associate with human factors. Furthermore, evaluating the 40-year data path established the foundational data trend required to develop and deploy a predictive model for the upcoming decade.

3.4 Predictive Modelling Framework

To project GHG concentration for the succeeding decade, a linear regression model was used. This was selected as an introductory predictive tool which supported the project's aim to develop a predictive model to analyse trends for the upcoming decade. Utilising this was decided after ANOVA test results indicated consistent growth trends, making the data suitable for predictive modelling. Linear regression provided an appropriate method to capture this trajectory, enabling a clear estimation of future forecast over the upcoming decade.

Prior to executing linear regression core assumptions were validated. The assumptions of linearity, spread and constant variance were evaluated using fitted vs residual plots, while the normality was confirmed using QQ-plots. This ensured reliability of forecast generated and provided with a foundational predictive model required for the scope of this study.

4.0 Results and Discussion

4.1 Univariate Analysis

Univariate analysis determined the baseline distribution of concentration levels of CO_2 and CH_4 recorded at Cape Grim (1986-2025) to develop an individual profile for each gas prior to identifying their correlation and trends. The five-number summary of CO_2 in Table 2.0, revealed a range from 342.29 to 423.28 ppm, with 376.05 ppm median. Its histogram shown in Figure 3.0 displayed a multimodal, right skewed distribution, capturing the natural seasonal variation upon the long-term upwards trend. Conversely, CH_4 concentration spanned from 1581.97 to 1894.99 ppb with 1735.50 ppb median as seen in Table 3.0. The histogram shown in Figure 4.0, depicted a symmetrical bell-shaped pattern. Nevertheless, a slight rightward tilt highlighted a recent increase in methane readings. Boxplots for both gases confirmed no outliers, validating dataset integrity as illustrated in Figure 5.0 and Figure 6.0.

Table 2.0: five-number summary of the carbon dioxide (CO_2) dataset

Minimum	Q1	Median	Q3	Maximum
342.289	357.611	376.053	397.183	423.279

Table 3.0: five-number summary of the methane (CH_4) dataset

Minimum	Q1	Median	Q3	Maximum
1581.970	1698.273	1735.500	1781.295	1894.990

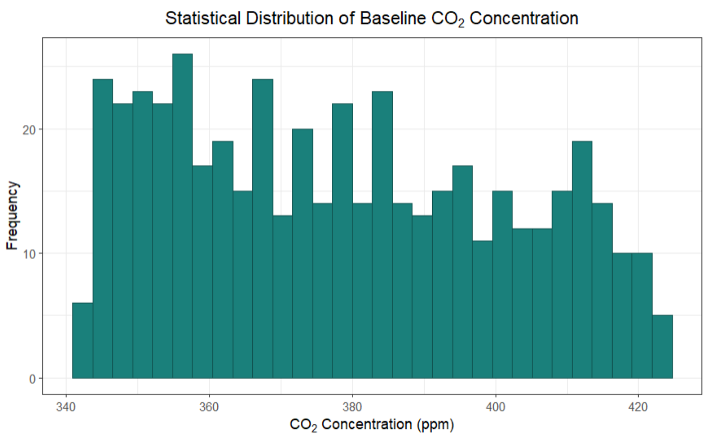

Figure 3.0: Histogram depicting how CO_2 concentration levels are distributed

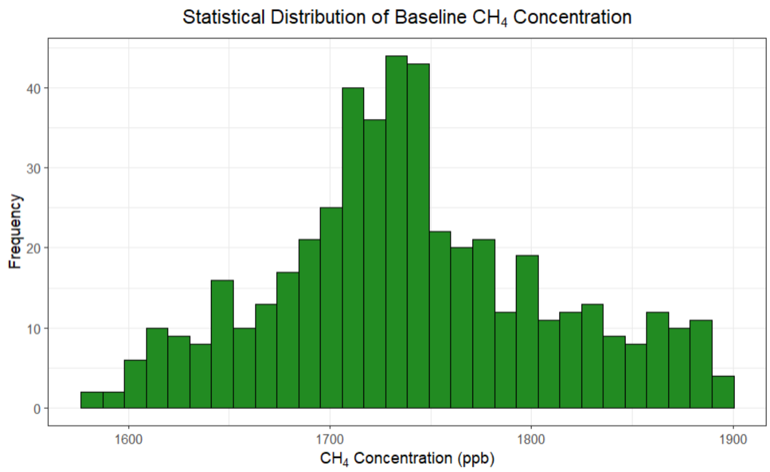

Figure 4.0: Histogram depicting how CH_4 concentration levels are distributed

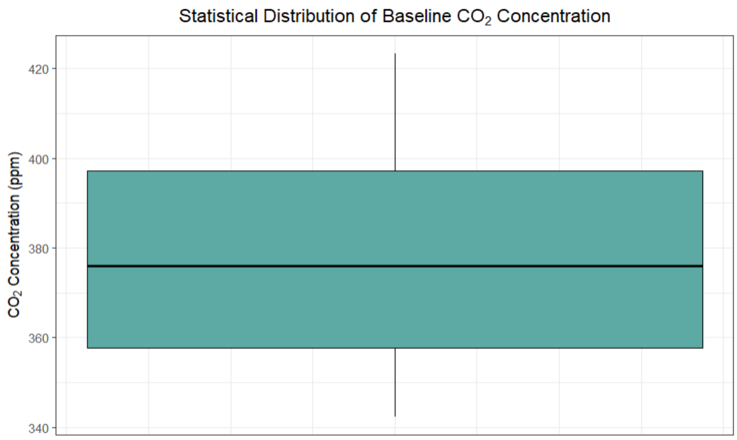

Figure 5.0: Boxplot illustrating the spread, range and outliers of CO_2 concentration levels

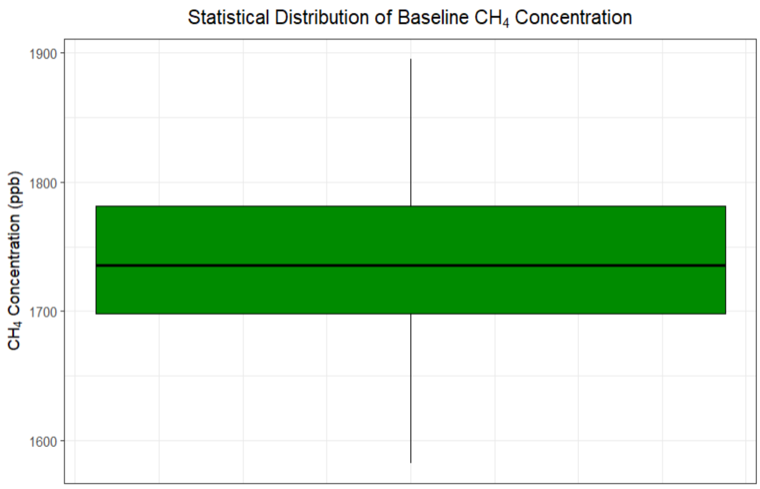

Figure 6.0: Boxplot illustrating the spread, range and outliers of CH_4 concentration levels

4.2 Bivariate Analysis

Bivariate analysis confirmed a very strong, positive linear correlation between atmospheric CO_2 and CH_4 concentrations, yielding Pearson correlation coefficient of r = 0.97. However, while a strong statistical association exists, it does not imply that both share the exact same drivers, as each gas responds to distinct anthropogenic and natural sources (Fang et al., 2020).

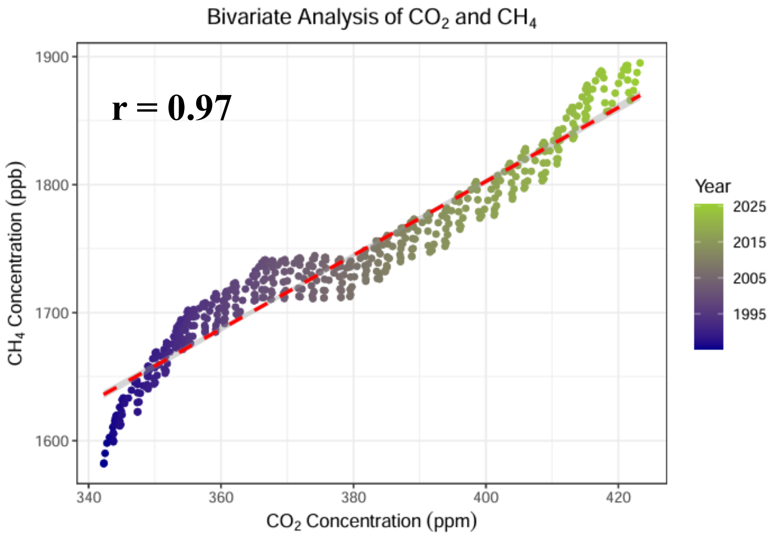

Figure 7.0: Bivariate analysis correlation of CO_2 and CH_4 at Cape Grim (1995-2025)

As depicted in Figure 7.0, a detailed inspection of the bivariate scatter profile revealed a non-linear deviation between 2005 and 2015. During this decade, the relationship exhibited a plateau: CO_2 concentrations continued their steep ascent, while CH_4 growth decelerated significantly, resulting in a temporary separation. Post 2015, the trajectory re-aligned as CH_4 growth accelerated rapidly, matching the upward momentum of CO_2. Further investigation will examine if the growth rates of these two gases remain identical over the long term.

4.3 ANOVA Test

ANOVA test was conducted to evaluate mean growth rates of CO_2 and CH_4 over the past decades. The hypotheses stated are depicted in Table 4.0.

Table 4.0: Hypotheses stated for ANOVA test

Null hypothesis (H_0)	The mean growth rate is equal across all four decades
Alternative hypothesis ($H_{a)}$)	At least one mean growth rate is different across all four decades.

The reliability of this analysis is verified by validating three assumptions. Firstly, the normality is confirmed via QQ-plots, as depicted in Figure 9.0 and Figure 10.0. For CO_2, the points follow the reference line, deeming it normally distributed, whilst CH4 indicates deviations. Nevertheless, the central limit theorem confirms normality for CH4, as the sample size is large ($n \approx 500$). Subsequently, constant variance (r) was calculated using formula shown in Figure 8.0. CO2 was 1.068, and CH4 was 1.160, both falling below the threshold. In terms of independence, all variables are independent in and between decades, as each measurement was captured from new air flow and further separated into different decades.

$$r = \frac{\max \mathrm{SD}}{\min \mathrm{SD}} < 2$$

Figure 8.0: Constant variance assumption for ANOVA test

Based on the results, H_o rejected for CO_2 ($p<0.05$) as p=0.04 and H_a retained for CH_4($p>0.05$) as p=0.77. Additionally, Tukey HSD shown in Table 5.0 illustrates that, although adjacent decades have not differentiation significantly, the variation between first and recent decade shows increase. Moreover, the side-by-side boxplot for CO_2 and CH_4 is depicted in Figure 11.0 and Figure 12.0 respectfully, further indicates that CO_2 increased and CH_4, was stable. Ultimately, these findings verify long-term expansion of both greenhouse gases.

Table 5.0: Tukey HSD of CO_2 imported from R for ANOVA test

Decades	Difference	lower	upper	p-adj
1996–2005 – 1986–1995	0.2183777	-0.67595948	1.112715	0.9225200
2006–2015 – 1986–1995	0.4823349	-0.41200228	1.376672	0.5059229
2016–2025 – 1986–1995	0.9595230	0.05551704	1.863529	0.0325555
2006–2015 – 1996–2005	0.2639572	-0.63037998	1.158294	0.8719354
2016–2025 – 1996–2005	0.7411453	-0.16286066	1.645151	0.1501833
2016–2025 – 2006–2015	0.4771881	-0.42681786	1.381194	0.5245831

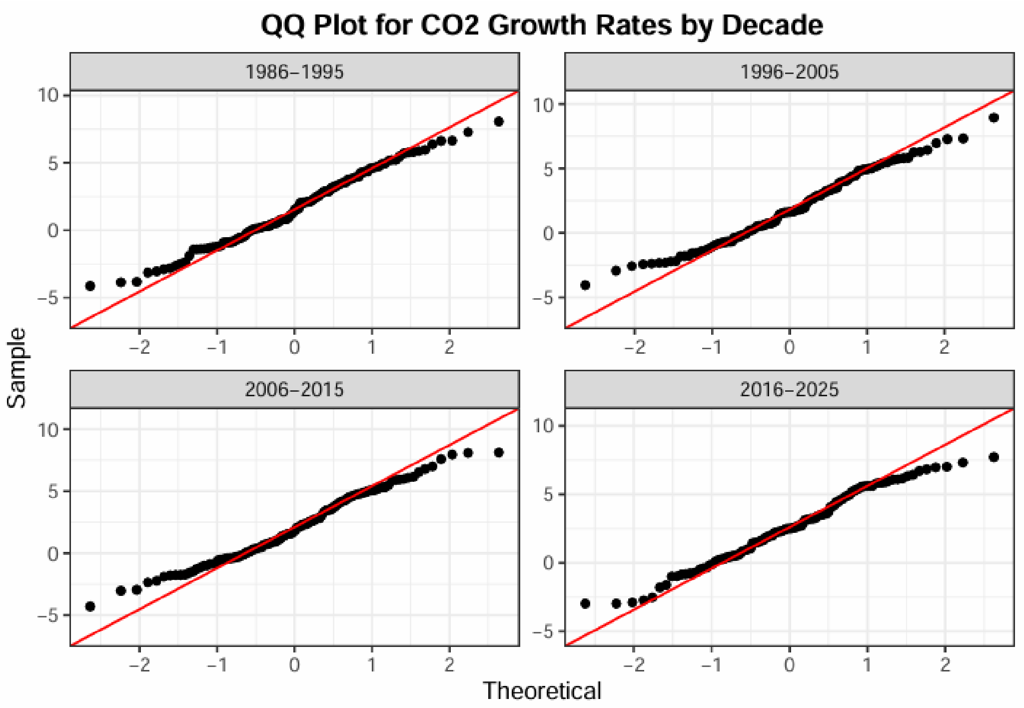

Figure 9.0: QQ Plots illustrating the growth rates of CO_2 by decades

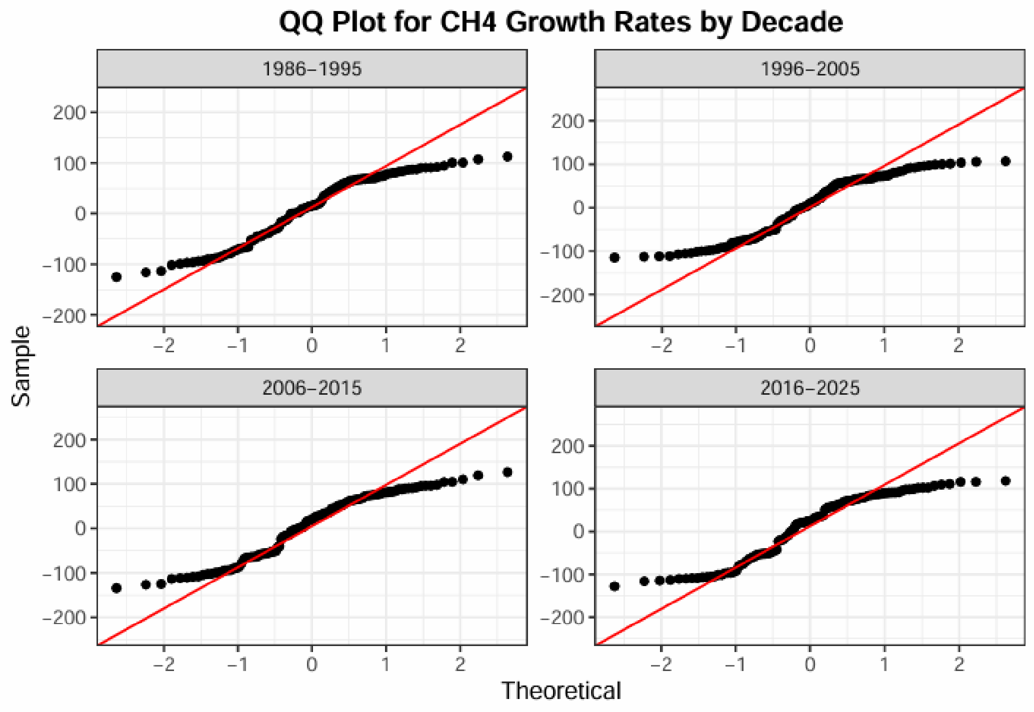

Figure 10.0: QQ Plots illustrating the growth rates of CH_4 by decades

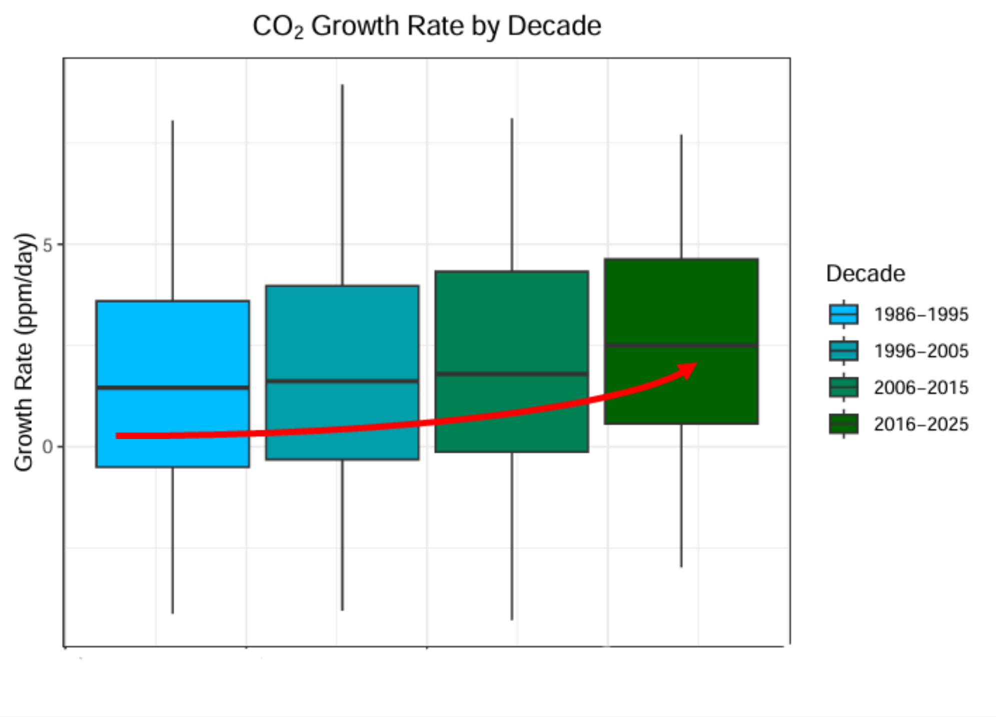

Figure 11.0: side-by-side boxplot of CO_2 Concentration growth rates for different decades

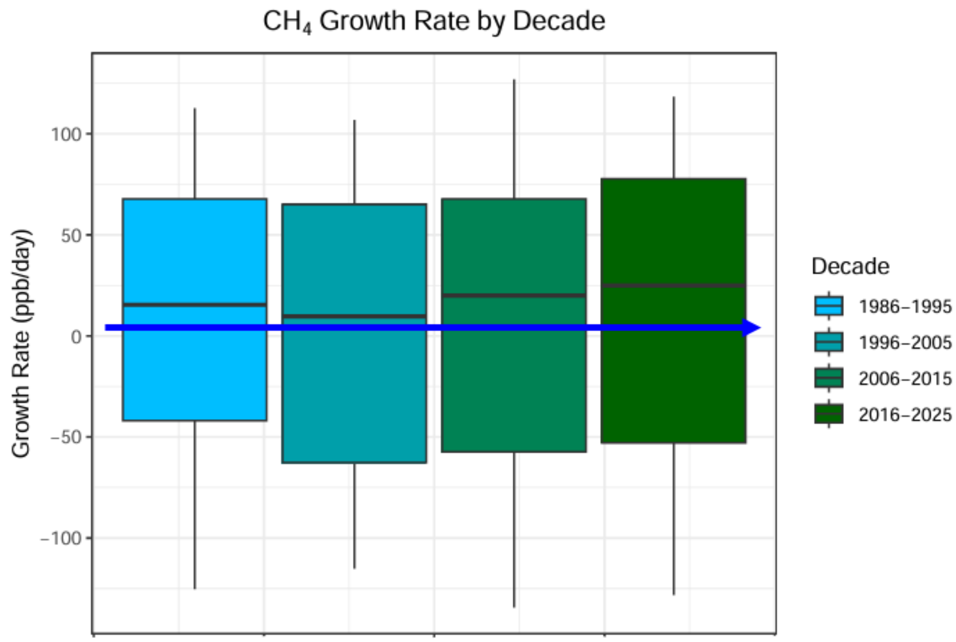

Figure 12.0: side-by-side boxplot of CH_4 Concentration growth rates for different decades

4.4 Linear Regression

Both models disobeyed the assumptions checking; however, linear regression was still used as an introductory predictive model to capture the overall trend of each gas, as seen in Figures 13.0 to 18.0. The CO_2 model illustrated a slope of 1.957 ppm per year, meaning it follows a very predictable upward trend. The CH_4 model showed a slope of 5.637 ppb per year, which is slightly less predictable. Both gases are increasing over time. However, CO_2 is accumulating consistently and confidently within the atmosphere. Their causes are also different. CO_2 is mainly emitted by industrial activity and burning fossil fuels. CH_4 mostly comes from agriculture and natural gas. Since CO_2 is rising faster and more predictably, reducing industrial emissions should be the priority for a mitigation strategy.

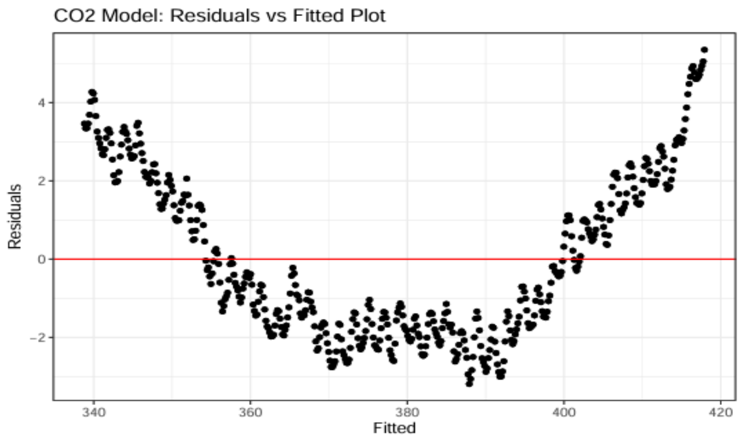

Figure 13.0: Linearity and Constant Spread based on Residuals vs Fitted Plot for CO_2

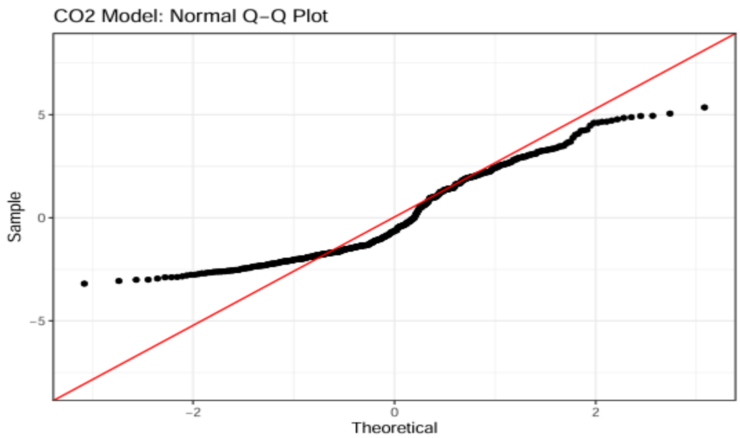

Figure 14.0: Assumption checking for CO_2 Normality using Q-Q plot

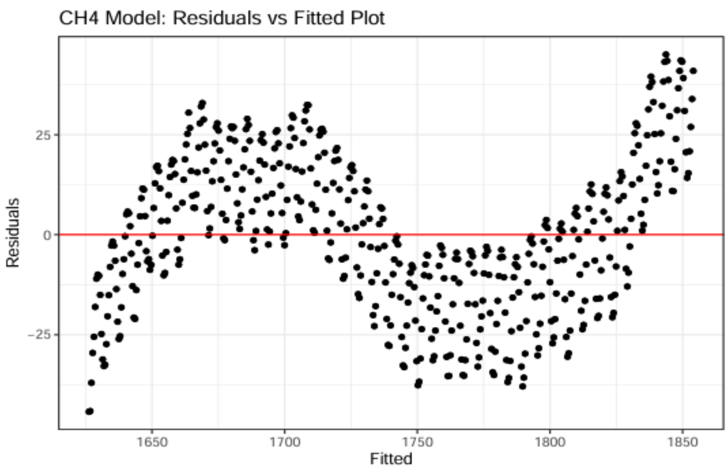

Figure 15.0: Linearity and Constant Spread based on Residuals vs Fitted Plot for CH_4

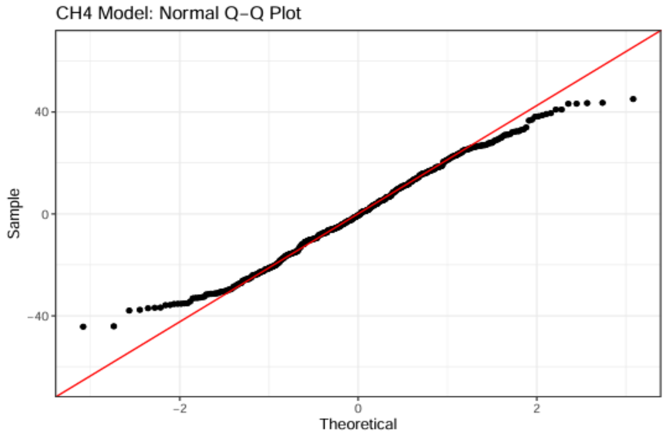

Figure 16.0: Assumption checking for CH_4 Normality using Q-Q plot

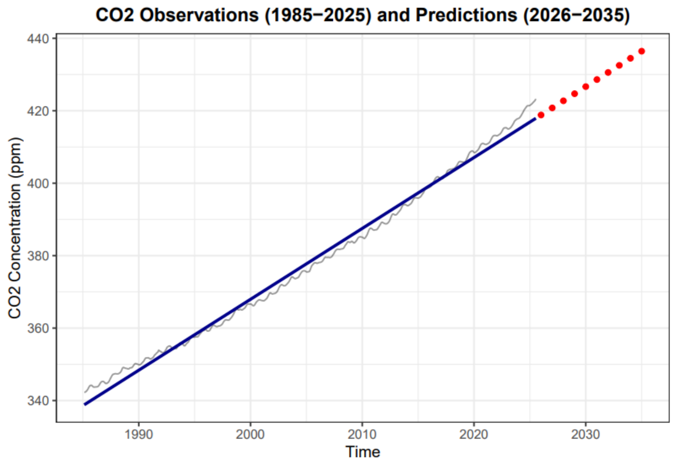

Figure 17.0: CO_2 observations from 1985-2025 and predictions for 2026-2035

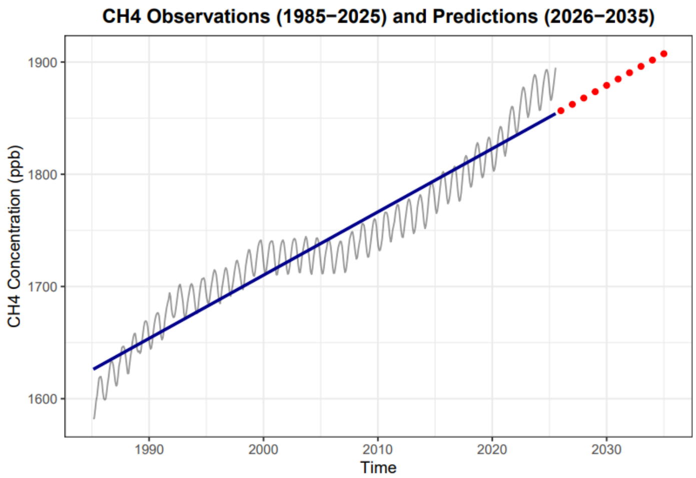

Figure 18.0: CH_4 observations from 1985-2025 and predictions for 2026-2035

5.0 Conclusion

This study investigated four decades of CO_2 and CH_4 atmospheric concentrations from Cape Grim to evaluate their correlation, measure long-term trends, and develop a predictive model for 2026–2035. Univariate analysis confirmed a reliable dataset, bivariate analysis identified a strong correlation, and ANOVA revealed that CO_2 growth rates accelerated significantly over decades compared to CH_4. Linear regression confirmed both gases will continue accumulating through 2035. Although CO_2 and CH_4 are correlated, their emission sources differ, and CO_2 is accumulating at a consistent rate atmospherically. Therefore, mitigation strategies must prioritise reducing industrialization and moving toward renewable energy sources. Predictive accuracy could be further improved by adopting an Autoregressive Integrated Moving Average (ARIMA) model, which utilises past data patterns, performs error correction, and stabilises data for more reliable forecasting.

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